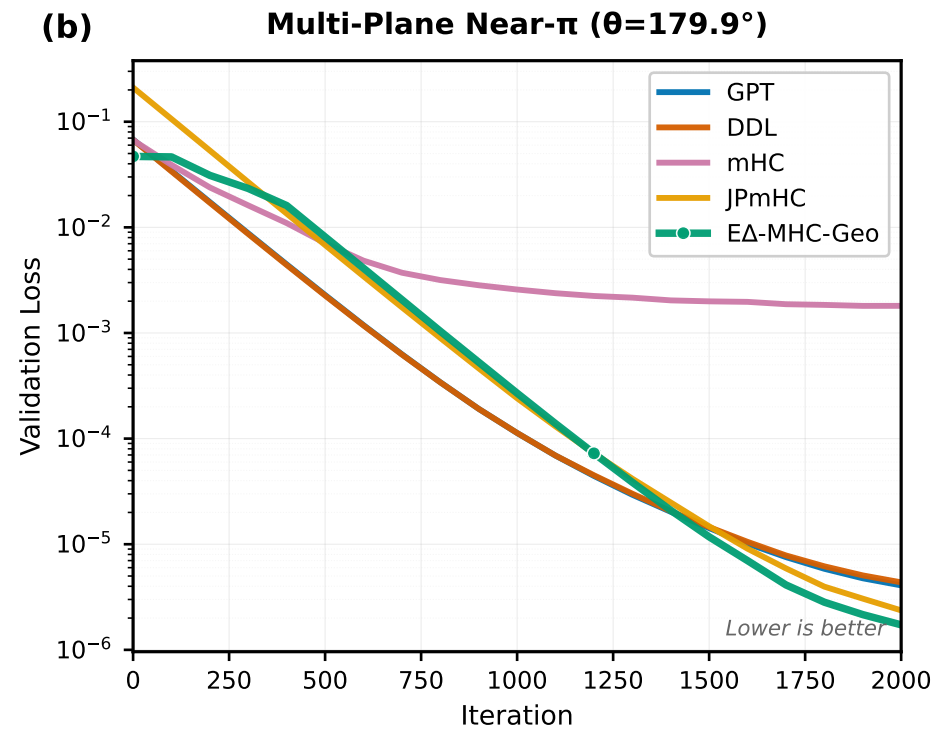
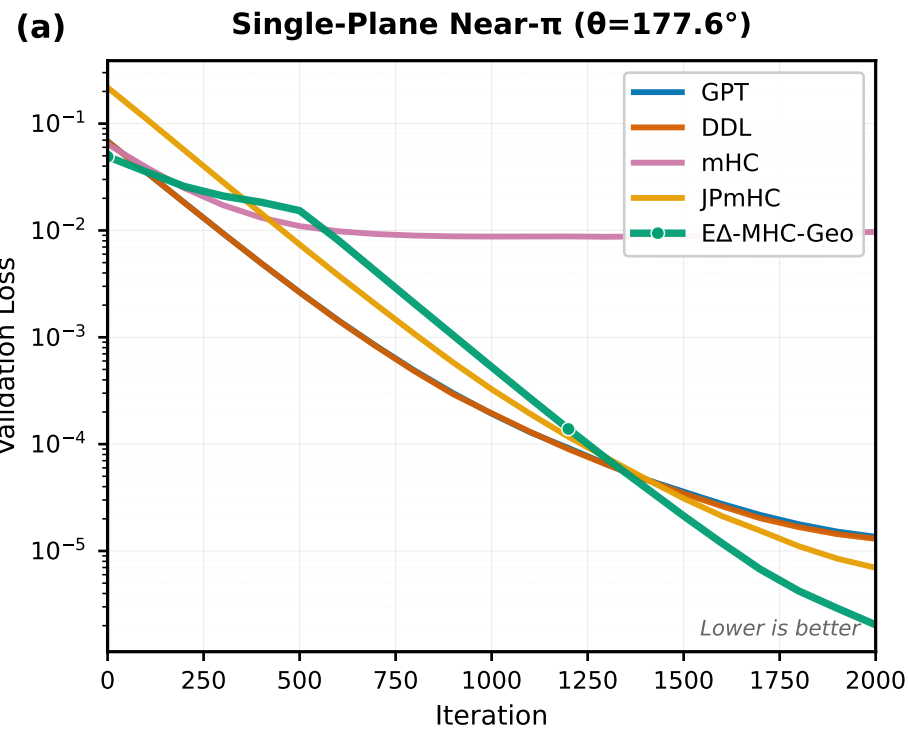


Near- π Rotation Analysis: Training Curves & Per-Layer Gate Adaptation



— Performance Summary —

Final Validation Loss:

Model	Single-Pl	Multi-Pl
GPT	1.20e-05	3.40e-06
DDL	1.16e-05	3.64e-06
mHC	9.83e-03	1.81e-03
JPmHC	5.45e-06	1.68e-06
EΔ	1.16e-06	1.29e-06

All $S0(n)$ models excel at near- π rotation (no negation).

